

PDC Target-Momentum Reconstruction: Algorithm, Physics, and Uncertainty

Current Status Report — April 2026

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2026-04-16

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Coordinate frames: lab vs. target (beam-as-Z)

Two right-handed frames are used consistently throughout this report:

- **Lab frame** (Geant4 world): $+z$ along the nominal SAMURAI beamline, $+y$ vertical up. PDC hits, field map, RK integration, and the internal least-squares fit all live here.
- **Target frame** (“beam-as-Z”, event-generator input): $+z$ along the incident beam direction after the $\alpha_{\text{tgt}} = 3^\circ$ target tilt; differs from lab by $R_y(-\alpha_{\text{tgt}})$. `truth_proton_p4` stays in this frame (`fMomentum` is never rewritten; the forward $R_y(-\alpha)$ only rotates a local copy used for the particle gun).

All truth vs. reco comparisons and all plots in this report are in the target frame. The reconstruction tools (`main.cc`, `analyze_pdc_rk_error.cc`) apply $R_y(+\alpha_{\text{tgt}})$ at the write stage; the 3×3 momentum covariance is rotated as $\Sigma' = R\Sigma R^T$ and $p_x/p_y/p_z$ intervals are rebuilt as Gaussian fallbacks from the rotated diagonals. $|\vec{p}|$ is rotation-invariant and its interval is passed through unchanged.

Prior to 2026-04-19, this rotation was missing and a systematic $\Delta p_x \approx -p_z \sin \alpha \approx -33 \text{ MeV}/c$ contaminated the residuals ($p_z \approx 627 \text{ MeV}/c$; if $p_z \approx 1 \text{ GeV}/c$ the offset is $-52 \text{ MeV}/c$). Fisher 68% coverage on p_x went from ≈ 0.05 back to ≈ 0.80 after the fix. All plots here were regenerated after the fix. Shared helpers: `libs/analysis_pdc_reco/include/PDCFrameRotation.hh`.

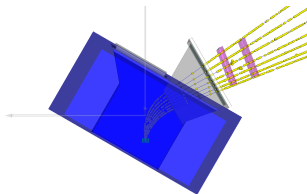
SAMURAI Spectrometer Geometry

Coordinate system: beam along $+z$, vertical y , horizontal x .

Key elements

- Target at $z \approx -1070$ mm (inside magnet)
- SAMURAI dipole: $B \approx 1.15$ T, center at origin, rotated 30° about y
- PDC1 at $z \approx 4000$ mm (after magnet)
- PDC2 at $z \approx 5000$ mm
- PDC angular tilt: $\theta_{\text{PDC}} = 57^\circ$

Typical proton: $|\vec{p}| \approx 635$ MeV/ c , $p_z \approx 627$ MeV/ c .



Geant4 VRML render (xz plane)

PDC Wire Coordinate System

PDC detects two 1D coordinates per plane: u and v .

Given the PDC angular tilt θ ($= 57^\circ$), the lab-frame wire directions are:

$$\hat{u} = \frac{1}{\sqrt{2}}(\cos \theta, +1, \sin \theta), \quad \hat{v} = \frac{1}{\sqrt{2}}(\cos \theta, -1, \sin \theta). \quad (1)$$

Wire coordinate values from the simulation (PDCSimAna):

$$u = \frac{x_{\text{rot}} + y}{\sqrt{2}}, \quad v = \frac{-x_{\text{rot}} + y}{\sqrt{2}},$$

where $x_{\text{rot}} = \Delta x \cos \theta - \Delta z \sin \theta$ is the position relative to PDC center, rotated by the PDC tilt angle.

Key point: $\hat{u}$ and $\hat{v}$ have non-zero z components. The z -component carries the curvature information from the magnetic deflection. Discarding it (projecting to the xy -plane) destroys the momentum sensitivity.

Relativistic Equation of Motion

A charged particle with charge q , momentum $\vec{p}$, and energy $E = \sqrt{|\vec{p}|^2 + m^2}$ in a magnetic field $\vec{B}(\vec{r})$ obeys:

$$\frac{d\vec{r}}{dt} = \frac{\vec{p} c^2}{E} = \vec{v}, \quad (2)$$

$$\frac{d\vec{p}}{dt} = q \vec{v} \times \vec{B}(\vec{r}). \quad (3)$$

Properties:

- The Lorentz force is perpendicular to $\vec{v}$, so $|\vec{p}|$ is conserved (no dE/dx in current model).
- In the rotated SAMURAI field (B_y -dominated), the primary bending occurs in the xz -plane.

Unit system: positions in mm, time in ns, momenta in MeV/ c , fields in T.

Conversion constant: $q_e \cdot c = 89.876 \text{ MeV}/(c \cdot \text{T} \cdot \text{ns})$.

4th-Order Runge-Kutta Integration (RK4)

State vector $\vec{Y} = (\vec{r}, \vec{p})$ is evolved with fixed step Δt :

$$\vec{K}_1 = f(\vec{Y}_n),$$

$$\vec{K}_2 = f(\vec{Y}_n + \frac{\Delta t}{2} \vec{K}_1),$$

$$\vec{K}_3 = f(\vec{Y}_n + \frac{\Delta t}{2} \vec{K}_2),$$

$$\vec{K}_4 = f(\vec{Y}_n + \Delta t \vec{K}_3),$$

$$\vec{Y}_{n+1} = \vec{Y}_n + \frac{\Delta t}{6} (\vec{K}_1 + 2\vec{K}_2 + 2\vec{K}_3 + \vec{K}_4),$$

where $f(\vec{r}, \vec{p}) = (\vec{p} c^2 / E, q \vec{v} \times \vec{B}(\vec{r}))$.

Each step queries the field map $\vec{B}(\vec{r})$ four times. The trajectory is stored as a sequence of $(\vec{r}_i, \vec{p}_i, t_i)$ points.

Default step: $\Delta t \sim 0.1$ ns (≈ 30 mm spatial step), ~ 150 – 200 steps from target to PDC2.

Parameterization

State vector (5 parameters):

$$\vec{\alpha} = (\delta x, \delta y, u, v, p),$$

where:

- $\delta x, \delta y$ — target position offset from nominal [mm]
- $u = p_x/p_z, v = p_y/p_z$ — direction tangents at the target
- $p = |\vec{p}|$ — total momentum magnitude [MeV/c]

Momentum vector reconstructed as:

$$p_z = \frac{p}{\sqrt{1 + u^2 + v^2}}, \quad p_x = u \cdot p_z, \quad p_y = v \cdot p_z.$$

Fit modes:

- kFixedTargetPdcOnly: 3 free params (u, v, p), $\delta x = \delta y = 0$ (fixed target)
- kThreePointFree: 5 free params ($\delta x, \delta y, u, v, p$) with soft target prior

Residual Construction

For each PDC plane $k \in \{1, 2\}$, the trajectory point closest to the measured hit $\vec{r}_k^{\text{meas}}$ is found. The 3D displacement is projected onto the wire directions:

$$r_u^{(k)} = (\vec{r}_k^{\text{traj}} - \vec{r}_k^{\text{meas}}) \cdot \hat{u}, \quad r_v^{(k)} = (\vec{r}_k^{\text{traj}} - \vec{r}_k^{\text{meas}}) \cdot \hat{v}. \quad (4)$$

These raw residuals are **whitened** via a 2×2 Cholesky decomposition of the measurement covariance:

$$C = \begin{pmatrix} \sigma_u^2 & \rho \sigma_u \sigma_v \\ \rho \sigma_u \sigma_v & \sigma_v^2 \end{pmatrix}, \quad \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = L^{-1} \begin{pmatrix} r_u \\ r_v \end{pmatrix}, \quad C = LL^T.$$

After whitening, each $w_i \sim \mathcal{N}(0, 1)$ is an independent, unit-variance residual.

Total: 4 residuals (2 per plane). In `kThreePointFree` mode, 2 additional target-constraint residuals: $w_5 = \delta x / \sigma_{\text{tgt}}$, $w_6 = \delta y / \sigma_{\text{tgt}}$.

χ^2 and Degrees of Freedom

$$\chi^2(\vec{\alpha}) = \sum_{i=1}^{N_{\text{res}}} w_i^2(\vec{\alpha})$$

Mode	Free params	PDC residuals	Target residuals	N_{res}	N_{dof}
kFixedTargetPdcOnly	u, v, p	4	0	4	1
kThreePointFree	$\delta x, \delta y, u, v, p$	4	2	6	1

The Jacobian $J_{ij} = \partial w_i / \partial \alpha_j$ is computed by central finite differences (each column requires two additional RK trajectory evaluations).

Levenberg-Marquardt Optimization

At each iteration, the normal equation is solved:

$$(J^T J + \lambda \cdot \text{diag}(J^T J)) \Delta \vec{\alpha} = -J^T \vec{w},$$

where λ is the damping parameter.

Update rules:

- If $\chi_{\text{new}}^2 < \chi_{\text{old}}^2$: accept step, $\lambda \rightarrow \lambda/3$
- Otherwise: reject step, $\lambda \rightarrow 4\lambda$

Convergence: $\Delta \chi^2 < 10^{-8}$ or $\|\Delta \vec{\alpha}\| < 10^{-5}$.

Initialization challenge: the straight-line direction from target to PDC is off by $\sim 25^\circ$ due to magnetic bending. Without a good starting point, LM converges to wrong local minima.

Coarse Grid Search + Multi-Start Strategy

Problem: single initial guess leads to convergence failures for many momentum values.

Solution: two-phase initialization.

Phase 1: Grid search over $N_u \times N_v \times N_p = 7 \times 3 \times 5 = 105$ candidate states:

- $u \in [u_{\text{line}} - 1.0, u_{\text{line}} + 1.0]$ (7 points, accounts for $\pm 45^\circ$ deflection)
- $v \in [v_{\text{line}} - 0.3, v_{\text{line}} + 0.3]$ (3 points, less out-of-plane bending)
- $p \in \{200, 400, 700, 1200, 2000\}$ MeV/ c

Each candidate is evaluated once (~ 0.1 ms per Evaluate). Total: ~ 10 ms.

Phase 2: Multi-start LM from the 5 lowest- χ^2 grid candidates. The LM result with the global-minimum χ^2 is selected.

Cost: $5 \times$ LM convergence ≈ 50 ms. Fully acceptable for offline reconstruction.

Overview of Error Analysis Methods

Method	Construction	Meaning	Cost
Fisher	$\Sigma = (J^T J)^{-1}$ at MLE	Local Gaussian $\approx$ of likelihood	1 Jacobian
Profile LH	Scan one α_k , minimize others; $\Delta\chi^2 = 1$ crossings	Frequentist CI	~ 20 LM per param
Laplace	$(J^T J + \Lambda_{\text{prior}})^{-1}$ at MAP	Bayesian credible interval (Gaussian $\approx$)	1 matrix inversion
MCMC	Metropolis-Hastings random walk in $\vec{\alpha}$ space	Full Bayesian posterior	$\sim 50k$ steps
MC Prop.	256 Gaussian samples in (u, v, p) , transform to (p_x, p_y, p_z)	Nonlinear error propagation	256 transforms

Fisher and Laplace are fast (every event); Profile LH and MCMC are offline diagnostics.
 MC propagation captures nonlinear asymmetry in the $(u, v, p) \rightarrow (p_x, p_y, p_z)$ mapping.

Run Configuration for Numerical Results

Subsequent frames in this section quote numbers from two complementary runs. The default RK fit mode is `kThreePointFree` (5 free params ($\delta x, \delta y, u, v, p$) with a soft target prior $\sigma_{xy}^{\text{tgt}} = 5$ mm); all examples here use `kFixedTargetPdcOnly` ($\delta x, \delta y$ fixed at 0; only (u, v, p) free), which matches the dataset prefixes shown below.

Source	Path	Use
Smoke (5 events)	<code>build/test_output/error_analysis_per_event</code>	Per-event Fisher table (frame “Error Analysis Results”)
Ensemble (744 events)	<code>build/test_output/ensemble_n50_target_frame_validation_cross_target_pdc_only</code>	Grid
Realistic legacy (815 tracks)	<code>results/pdc_rk_error_after_pdcanc_20260416_zh</code>	Report: per-track Fisher/Laplace + 32 Profile + 16 MCMC; no truth $\Rightarrow$ no coverage

The Chinese companion (`rk_reconstruction_status_20260416_zh.pdf`) contains a worked example, the per-method comparison plots from the realistic-data run, and a discussion of why MCMC is broader than Profile in that run.

Fisher Information

At the MLE $\hat{\vec{\alpha}}$, the observed Fisher information matrix is:

$$\mathcal{I}(\hat{\vec{\alpha}}) = J^T J,$$

where $J_{ij} = \partial w_i / \partial \alpha_j$ is the whitened-residual Jacobian.

The parameter covariance (Cramér-Rao bound) is:

$$\Sigma_{\vec{\alpha}} = \mathcal{I}^{-1} = (J^T J)^{-1}.$$

Momentum covariance is propagated via the momentum Jacobian $M_{ij} = \partial p_i / \partial \alpha_j$:

$$\Sigma_{\vec{p}} = M \Sigma_{\vec{\alpha}} M^T.$$

Fisher confidence intervals (assuming Gaussian):

$$p_i \pm z_{\alpha} \cdot \sqrt{(\Sigma_{\vec{p}})_{ii}}, \quad z_{0.68} \approx 1.0, \quad z_{0.95} \approx 1.96.$$

Fisher: Algorithm Outline and Cost

Per-event computation by `PDCErrorAnalysis::computeFisher`
(`libs/analysis_pdc_reco/src/PDCErrorAnalysis.cc`):

- 1 **Jacobian** $J = \partial \vec{w} / \partial \vec{\alpha}$ via central differences; $2n_{\text{params}}$ RK trajectory propagations, step $h_u = \max(10^{-4}, 0.01|u|)$, $h_p = \max(0.5 \text{ MeV}/c, 10^{-3}p)$.
- 2 **Information** $\mathcal{I} = J^T J$ (3×3 for `kFixedTargetPdcOnly`, 5×5 for `kThreePointFree`).
- 3 **SVD** $\mathcal{I} = USV^T$; report condition $\kappa = S_{\text{max}}/S_{\text{min}}$, fail if κ exceeds threshold.
- 4 **Param covariance** $\Sigma_{\vec{\alpha}} = VS^{-1}U^T$.
- 5 **Momentum Jacobian** $M = \partial(p_x, p_y, p_z) / \partial \vec{\alpha}$ (central differences).
- 6 **Momentum covariance** $\Sigma_{\vec{p}} = M \Sigma_{\vec{\alpha}} M^T$.

Cost: 1 Jacobian eval (~ 6 RK propagations) + 1 SVD ≈ 5 ms per event $\Rightarrow$ runs inline.

Worked example (event 0 from the smoke run, MLE point estimate
 $(p_x, p_y, p_z) = (-107.33, -13.35, 610.90) \text{ MeV}/c$):

$$\sigma_{p_x} = 3.83, \sigma_{p_y} = 0.15, \sigma_{p_z} = 2.22 \text{ MeV}/c, \quad \chi_{\text{red}}^2 = 0.18.$$

The order-of-magnitude difference between $\sigma_{p_y} \sim 0.15$ and $\sigma_{p_x}, \sigma_{p_z} \sim 2\text{--}4 \text{ MeV}/c$ is the geometric Jacobian effect: the wire direction has its largest projection on $\hat{y}$ ($\pm 1/\sqrt{2}$), so the wire residual most

Monte Carlo Nonlinear Error Propagation

The transformation $(u, v, p) \rightarrow (p_x, p_y, p_z)$ is nonlinear:

$$p_z = \frac{p}{\sqrt{1 + u^2 + v^2}}, \quad p_x = u \cdot p_z, \quad p_y = v \cdot p_z.$$

Procedure:

- 1 From Fisher covariance $\Sigma_{\vec{\alpha}}$, generate $N = 256$ samples: $\vec{\alpha}^{(k)} \sim \mathcal{N}(\hat{\vec{\alpha}}, \Sigma_{\vec{\alpha}})$.
- 2 Transform each sample to momentum space: $\vec{p}^{(k)} = \vec{p}(\vec{\alpha}^{(k)})$.
- 3 Compute empirical percentiles:
 - 68% interval: $[q_{16}, q_{84}]$
 - 95% interval: $[q_{2.5}, q_{97.5}]$

Advantages:

- Captures asymmetric intervals automatically.
- No linearization (delta method) assumption needed.
- Cheap: only 256 algebraic transforms, no RK trajectory evaluations.

Profile Likelihood

For each active parameter α_k , scan a grid of values while minimizing χ^2 over the remaining parameters:

$$\chi_{\text{profile}}^2(\alpha_k) = \min_{\alpha_{j \neq k}} \chi^2(\vec{\alpha}).$$

The confidence interval is defined by:

$$\Delta\chi^2 = \chi_{\text{profile}}^2(\alpha_k) - \chi_{\text{min}}^2 \leq \Delta\chi_{\text{crit}}^2,$$

where $\Delta\chi_{\text{crit}}^2 = 1.0$ for 68% CL and 4.0 for 95% CL (1 parameter of interest).

Implementation: warm-start LM (previous scan point provides the initial guess for the next), linear interpolation for crossing detection.

Use case: diagnose non-Gaussian/asymmetric likelihood structure. Especially important for p near the boundary (p_{min} or p_{max}).

Bayesian Posterior (Laplace + MCMC)

Posterior:

$$\log \pi(\vec{\alpha} \mid \text{data}) \propto -\frac{1}{2}\chi^2(\vec{\alpha}) + \log \pi_0(\vec{\alpha}),$$

where the prior π_0 is flat on $(u, v, \delta x, \delta y)$ and optionally Gaussian on p :

$$\log \pi_0(p) = -\frac{(p - p_0)^2}{2\sigma_p^2}.$$

Laplace approximation: Gaussian around MAP with covariance $(J^T J + \Lambda_{\text{prior}})^{-1}$.

MCMC: Metropolis-Hastings with adaptive proposal:

- Proposal: $\vec{\alpha}' = \vec{\alpha} + s \cdot L \cdot \vec{z}$, $\vec{z} \sim \mathcal{N}(0, I)$, $LL^T = \Sigma_{\text{posterior}}$
- Acceptance: $\min(1, \exp(\Delta \log \pi))$
- Adaptive scale s : adjusted during burn-in to target 23% acceptance rate
- Burn-in: 5000 steps; post-burn: 50000 steps, thin every 5
- Diagnostics: Geweke z-score (stationarity), effective sample size (autocorrelation)

E2E Test Results: RK vs NN [NN reference contaminated]

Audit note (2026-04-20): NN bias columns below were produced by the now-deleted polluted model `formal_B115T3deg_qmdwindow/20260227_223007/` (target-contamination audit: `docs/audit/2026-04-20-nn_target_contamination_audit.md`). Clean retrain has val p_x RMSE -51% , p_y -81% , p_z essentially unchanged — so the qualitative p_z comparison (RK -10.8 vs NN -3 MeV/c, dE/dx -dominated) still holds, but p_x, p_y “comparable” conclusion is invalidated. Re-run `test_rk_nn_comparison.py` with `formal_B115T3deg/20260420_184345/model/model.pt` before re quoting.

Test setup: Geant4 simulation $\rightarrow$ PDC reconstruction $\rightarrow$ target momentum reconstruction $\rightarrow$ truth comparison. 5 truth grid points, $p_z = 627$ MeV/c, 1 event each.

p_x^{true} (MeV/c)	RK bias (MeV/c)			NN bias (MeV/c)		
	Δp_x	Δp_y	Δp_z	Δp_x	Δp_y	Δp_z
-100	-7.3	-13.4	-16.1	-6.2	-13.0	-5.4
-50	-20.4	-25.5	-10.8	-20.5	-24.9	-0.7
0	3.3	-3.9	-13.1	3.5	-3.8	-3.0
+50	-33.3	16.3	-8.7	-31.4	16.8	-0.2
+100	-37.8	-20.9	-5.3	-43.6	-19.4	6.3

Three RK Modes — Unified Performance [NN row contaminated]

All three modes now use the same `RkLeastSquaresAnalyzer` with correct wire-coordinate residuals and multi-start initialization.

Mode	Mean $ \text{bias}_{p_x} $ (MeV/c)	Mean $ \text{bias}_{p_y} $ (MeV/c)	Mean $ \text{bias}_{p_z} $ (MeV/c)
kFixedTargetPdcOnly	20.4	16.0	10.8
kTwoPointBackprop	20.4	16.0	10.8
kThreePointFree	21.3	16.2	10.3
NN (reference, polluted model)	21.0	15.6	3.1

Before the bug fixes, the RK modes had mean biases of 180–650 MeV/c on p_x and 190–605 MeV/c on p_z . The improvement is **10–80 $\times$** . NN reference row pending re-eval with clean model; RK numbers are geometry-only and unaffected.

Cramér-Rao Bound from Spectrometer Geometry

Bending radius and deflection:

$$B\rho = \frac{p}{0.3q} = \frac{635}{300} = 2.12 \text{ T}\cdot\text{m}, \quad \rho = \frac{B\rho}{B} = \frac{2.12}{1.15} = 1840 \text{ mm},$$

$$\theta_{\text{bend}} \approx \frac{L_{\text{eff}}}{\rho} = \frac{800}{1840} \approx 0.43 \text{ rad} \approx 25^\circ.$$

Momentum resolution from two-point exit angle measurement:

$$\sigma_\theta = \frac{\sigma_x \sqrt{2}}{L_{12}}, \quad \sigma_p = \frac{\sigma_\theta}{|\partial\theta_{\text{bend}}/\partial p|} = \frac{p}{\theta_{\text{bend}}} \cdot \sigma_\theta.$$

σ_x	σ_θ (mrad)	σ_p (MeV/c)	Note
0.5 mm	0.71	1.0	Geant4 smearing
2.0 mm	2.83	4.1	Fitter setting

This is the *position-only* limit assuming no multiple scattering.

Multiple Scattering: Highland Formula

For a 635 MeV/c proton ($\beta = 0.560$, $\beta p = 356$ MeV/c):

$$\theta_0 = \frac{13.6 \text{ MeV}}{\beta p} \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln(x/X_0) \right].$$

Material	Thickness	X_0 (g/cm ²)	x/X_0	θ_0 (mrad)
Sn target	2.5 mm (avg)	8.82	0.207	16.4
Air (1 atm)	1 m	36.66	3.29×10^{-3}	1.72
PDC gas [†]	224 mm	24.1	1.74×10^{-3}	1.21

[†]75% Ar + 25% i-C₄H₁₀ at 1 atm; effective path at 32° incidence; per PDC.

Target (16.4 mrad): absorbed by direction fit parameters (u, v); negligible effect on $|\vec{p}|$.

Air + PDC gas: scattering *after* the field corrupts the exit angle measurement.

Combined Resolution Estimate

MS corrupts the exit angle by $\sigma_{\theta, \text{MS}} = \sqrt{\theta_{0, \text{air}}^2 + \theta_{0, \text{PDC1}}^2} = \sqrt{1.72^2 + 1.21^2} = 2.10 \text{ mrad}$.

Using $\sigma_p = (p/\theta_{\text{bend}}) \sigma_\theta$:

Source	σ_θ (mrad)	σ_p (MeV/c)
Position ($\sigma_x=0.5 \text{ mm}$)	0.71	1.0
MS: Air (1 m)	1.72	2.5
MS: PDC1 gas	1.21	1.8
Combined ($\sigma_x=0.5 \text{ mm}$)	2.22	3.2
Combined ($\sigma_x=2.0 \text{ mm}$)	3.52	5.1

	Theory ($\sigma_x=0.5$)	NN MAE (p_z)	RK MAE (p_z)
p_z	3.2 MeV/c	3.1 MeV/c	10.8 MeV/c

NN/RK MAE from `test_rk_nn_comparison.py` on 5 E2E grid events. RK p_z bias (-10.8 MeV/c mean) attributed to missing dE/dx . **NN p_z from polluted model; however target contamination perturbed p_x, p_y much more than p_z , so p_z comparison stays qualitatively intact.**

Error Analysis Results (from analyze_pdc_rk_error)

All numbers from actual test output — build/test_output/error_analysis_rk/.

Event	$\sigma_{p_x}^{\text{Fish}}$ (MeV/c)	$\sigma_{p_y}^{\text{Fish}}$ (MeV/c)	$\sigma_{p_z}^{\text{Fish}}$ (MeV/c)	χ_{red}^2
0	3.83	0.15	2.22	0.178
1	3.20	0.16	2.05	1.896
2	6.50	0.15	5.41	0.613
3	6.48	0.15	5.64	0.065
4	6.96	0.20	6.80	0.696

Metric	Value (MeV/c)
Median Fisher $ \vec{p} $ width (68%)	10.72
Median Profile $ \vec{p} $ width (68%)	10.42
Median MCMC $ \vec{p} $ width (68%)	11.33
Median MCMC acceptance rate	0.51

Point estimates from 5 E2E test events; coverage validated separately on a 744-event ensemble (next frame) to

Ensemble Coverage Validation — Setup & Source

Dataset: 15 truth points ($p_x \in \{-100, -50, 0, 50, 100\}$, $p_y \in \{-20, 0, 20\}$ MeV/c, $p_z = 627$), **50 independent Geant4 replicas** each $\rightarrow$ 750 events; **744 matched** (99.2%). The Sn target is *disabled* in this geometry: the tree-gun injects protons directly at the target z-position, so leaving the 5 mm Sn slab in would only add spurious multiple-scattering that is absent from the real PDC tracking study. With the target removed, $p_y = \pm 50$ MeV/c deflects to ~ 310 mm at PDC2 — outside the tracker's y-acceptance — so the p_y grid is trimmed to ± 20 (tests/integration/test_pdc_target_momentum_e2e.cc:38). Fisher/Laplace on all 744; Profile on 32 χ^2 -stratified samples; MCMC on 64 (160 samples / 80 burn-in).

Reproduction (from smsimulator5.5/):

```
# 1. Build (requires anaroot-env)
micromamba activate anaroot-env && ./build.sh

# 2. Sim + reco + error analysis ( 40 min on 1 core)
ENSEMBLE_SIZE=50 PROFILE_PER_QUARTILE=8 MCMC_PER_QUARTILE=16 \
  OUT_ROOT=build/test_output/ensemble_n50 \
  bash scripts/analysis/run_ensemble_coverage.sh

# 3. Coverage aggregation (Clopper-Pearson CIs, pull stats, LaTeX)
python3 scripts/analysis/analyze_ensemble_coverage.py \
  --input-dir build/test_output/ensemble_n50/rk_fixed_target_pdc_only_error_analysis \
  --output-dir build/test_output/ensemble_n50/coverage_final
```

Ensemble Coverage — Per-Method Errors vs. True Residual

For each component we list *each method's 68% half-width* ($\pm$ in MeV/c; for Fisher/Laplace this is σ), the ratio to the robust residual half-width $\frac{1}{2}(p_{84}-p_{16})$ (ideal ≈ 1), and the observed 68% coverage. Robust metrics are used because a handful of LM failures give long tails that inflate the raw std:

Comp.	Med. bias	Res. hw	Method	Half-width (68%)	Half-w / Res. hw	Cov. 68%	CP 95% CI
p_x	-0.33	5.98	Fisher	7.40	1.24	0.801	[0.771, 0.829]
			Laplace	7.40	1.24	0.801	[0.771, 0.829]
			Profile	7.06	1.18	0.711	[0.624, 0.788]
p_y	-0.04	1.40	Fisher	0.75	0.54	0.421	[0.385, 0.457]
			Laplace	0.75	0.54	0.421	[0.385, 0.457]
			Profile	0.57	0.41	0.367	[0.284, 0.457]
p_z	-0.65	5.57	Fisher	6.94	1.25	0.772	[0.740, 0.801]
			Laplace	6.94	1.25	0.772	[0.740, 0.801]
			Profile	5.61	1.01	0.695	[0.608, 0.774]
$ \vec{p} $	-0.66	5.30	Fisher	6.68	1.26	0.761	[0.728, 0.791]
			Laplace	6.68	1.26	0.769	[0.737, 0.799]
			Profile	6.19	1.17	0.781	[0.700, 0.849]
			MCMC	5.25	0.99	0.633	[0.543, 0.716]

Takeaways (post 2026-04-19 coordinate-frame fix, target frame, all four methods re-run): (i) Fisher/Laplace half-widths for $|\vec{p}|$, p_x , p_y are within $\sim 25\%$ of the realised residual spread (ratio 1.24–1.26); Fisher 68% coverage lands near nominal

How the Four Methods Relate — Point Estimate vs. Interval

Key point: the *reco point estimate* is computed only once per event (LM best-fit on the RK trajectory). The four methods differ only in how they put an uncertainty interval *around* that same point.

Per event:

1. LM + RK4 integration $\rightarrow$ best-fit (p_x, p_y, p_z) minimizing χ^2
 $\rightarrow$ reco_px, reco_py, reco_pz in track_summary.csv (shared by all 4 methods)
2. Each method attaches its own 68%/95% interval to this point estimate:

Method	RK calls per event	Interval recipe
Fisher	1 (at best-fit; Jacobian)	$\hat{p} \pm \sigma$ from $(J^T J)^{-1}$; symmetric, Gaussian
Laplace	same as Fisher	same covariance matrix $\Rightarrow$ identical width
Profile	$\sim 17 \times 5$ per comp.	scan a parameter, refit others, find $\Delta\chi^2=1 / 4$; asymmetric OK
MCMC	~ 160 chain steps	Metropolis-Hastings on posterior $\rightarrow$ 16%/84% quantiles

Consequence: all four share the *same* forward model (RK4 trajectory + PDC residuals). A bias or linearization error in RK propagates into every interval. The 2026-04-19 coordinate-frame fix eliminated what looked like “persistent p_x bias -33 MeV/ c ”: reco output is now rotated from lab into target frame and the residual p_x median is -0.33 MeV/ c . Fisher/Laplace p_x coverage recovered from 0.05 to

What "Coverage" Means Here (Terminology)

Two different "CIs" in the table — don't confuse them:

- 1 **Method interval** (per-event): each method produces a 68%/95% interval $[lo_i, hi_i]$ for event i .
E.g. Fisher 68%: $\hat{p} \pm \sigma_{\text{Fisher}}$; MCMC 68%: 16%/84% quantiles of posterior samples.
- 2 **Coverage** (fraction, over the ensemble):

$$\text{cov} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[lo_i \leq p_i^{\text{truth}} \leq hi_i].$$

A well-calibrated method's 68% interval should cover truth with probability 0.68. Observed $< 0.68 \Rightarrow$ over-confident (intervals too narrow); observed $> 0.68 \Rightarrow$ over-conservative.

- 3 **Clopper-Pearson 95% CI** (last column of next table): the coverage *itself* is a binomial estimator with uncertainty $\propto 1/\sqrt{N}$. CP CI is the exact binomial confidence interval on that observed fraction — tells you whether the deviation from the nominal value is statistically significant.

Worked example: row 1 of the next table says Fisher $|\vec{p}|$ 68% coverage = 0.761, CP 95% CI = $[0.728, 0.791]$. That means $744 \times 0.761 = 566$ events out of 744 had truth inside Fisher's $\hat{p} \pm \sigma_{\text{Fisher}}$ window, and the nominal 0.68 sits just below the CP interval $\Rightarrow$ Fisher for $|\vec{p}|$ is *slightly* over-conservative (sampling noise alone cannot explain the excess).

Ensemble Coverage — Full Table

Coverage = fraction of 744 (or subset) events whose truth falls inside the method's stated interval. **CI column** = CP 95% binomial CI *on the coverage value itself* (source: `coverage_with_ci.csv`):

Comp.	Method	N	68%	68% CI	95%	95% CI
$ \vec{p} $	Fisher	744	0.761	[0.728, 0.791]	0.901	[0.877, 0.921]
$ \vec{p} $	Laplace	744	0.769	[0.737, 0.799]	—	—
$ \vec{p} $	MCMC	128	0.633	[0.543, 0.716]	0.898	[0.833, 0.945]
$ \vec{p} $	Profile	128	0.781	[0.700, 0.849]	0.930	[0.871, 0.967]
p_x	Fisher	744	0.801	[0.771, 0.829]	0.910	[0.887, 0.930]
p_x	Laplace	744	0.801	[0.771, 0.829]	—	—
p_x	Profile	128	0.711	[0.624, 0.788]	0.930	[0.871, 0.967]
p_y	Fisher	744	0.421	[0.385, 0.457]	0.690	[0.655, 0.723]
p_y	Laplace	744	0.421	[0.385, 0.457]	—	—
p_y	Profile	128	0.367	[0.284, 0.457]	0.625	[0.535, 0.709]
p_z	Fisher	744	0.772	[0.740, 0.801]	0.902	[0.878, 0.922]
p_z	Laplace	744	0.772	[0.740, 0.801]	—	—
p_z	Profile	128	0.695	[0.608, 0.774]	0.883	[0.814, 0.933]

Verdict (post 2026-04-19 rotation fix; all four methods re-run on post-fix data):

Step 1 — PDC Hit Dispersion under Identical Injection

Question. If you fire the same proton (same $\vec{p}_{\text{truth}}$) 50 times, how much does the PDC1/PDC2 hit position wander? With the Sn target disabled, this isolates the *residual* G4 stochasticity after the magnet: multiple scattering in the PDC gas (ArC2H6, 10 mm drift gap) plus PDC wire smearing ($\sigma_{\text{wire}} \sim 0.2$ mm) — *before* any reconstruction enters.

Data. For each of 5 special truth points we scatter-plot PDC1 (x,y) and PDC2 (x,y) from the $N=49\text{--}50$ matched reco events at that point. Global PDC1 is at $z \approx 1858$ mm and PDC2 at $z \approx 2301$ mm in SAMURAI coordinates; both are several meters downstream of the target/magnet — the PDC gas plus its 10 mm air gap dominate the remaining dispersion now that the 5 mm Sn slab is gone.

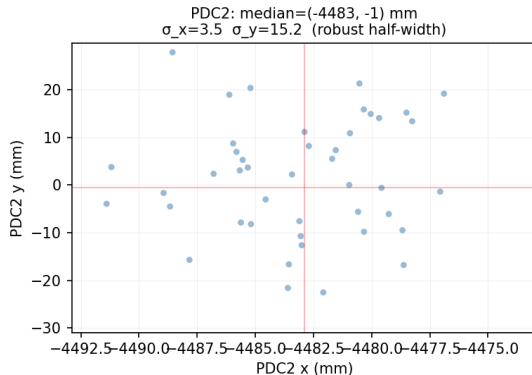
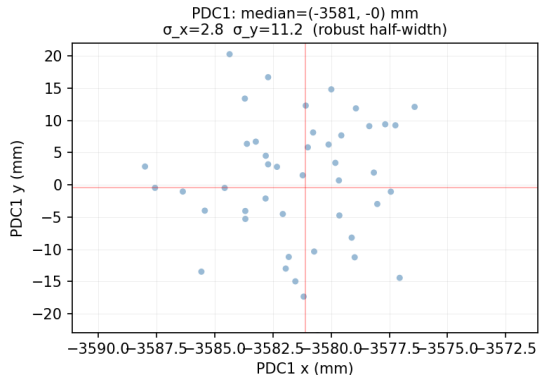
Source scripts:

- `scripts/analysis/extract_pdc_hits_from_reco.py` (PyROOT $\rightarrow$ CSV)
- `scripts/analysis/plot_pdc_hit_dispersion.py` (one PNG per truth point)
- Output:
`docs/reports/reconstruction/figures/g4_fluctuation/pdc_hits/px{...}_py{...}.png`

Key observation. Dispersion is strongly *anisotropic*: $\sigma_{\text{PDC},y} \gg \sigma_{\text{PDC},x}$ (factor 3–4). SAMURAI dipole bends in x-z, so the field constrains x but leaves y free to drift with every multiple-scattering kick. This is the geometric root cause of the p_y information collapse. Without the Sn target the absolute scale

PDC Hit Dispersion — Baseline (p_x, p_y) = (0, 0)

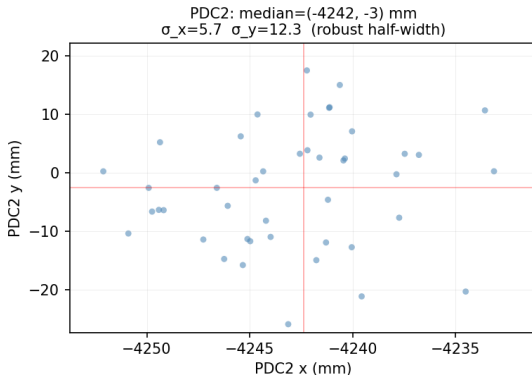
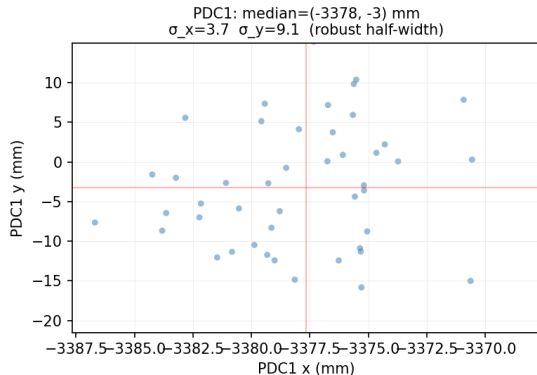
PDC hit dispersion: identical Geant4 injection (truth p_x, p_y) = (+0, +0) MeV/c, $p_z=627$, N=49



49 replicas of identical truth (0, 0, 627) MeV/c. PDC1: $\sigma_x=2.8$, $\sigma_y=11.2$ mm. PDC2: $\sigma_x=3.5$, $\sigma_y=15.2$ mm. $\sigma_y/\sigma_x \approx 4$. PDC2 spread is larger than PDC1 by $\sim 30\%$ — consistent with added drift length (PDC1 $\rightarrow$ PDC2 gas gap) scattering the already-dispersed trajectory further.

PDC Hit Dispersion — $(p_x, p_y) = (+100, 0)$

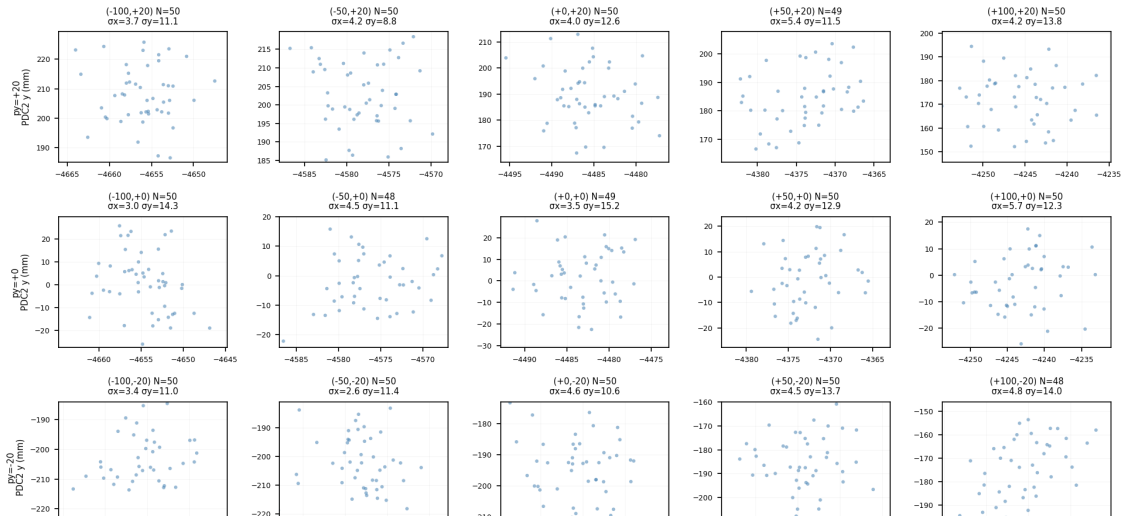
PDC hit dispersion: identical Geant4 injection (truth $p_x, p_y) = (+100, +0)$ MeV/c, $p_z=627$, $N=50$



50 replicas at $(+100, 0, 627)$. PDC1 $\sigma_x=3.7$, $\sigma_y=9.1$ mm. PDC2 $\sigma_x=5.7$, $\sigma_y=12.3$ mm. Compared with $(0, 0)$ baseline: σ_x grows modestly with $|p_x|$ (larger bend radius $\Rightarrow$ slightly longer in-field path), σ_y drops because the outgoing track crosses the PDC gas at a steeper angle (p_y/p_z projection reduced).

PDC2 Hit Dispersion — Full 5×3 Truth Grid

PDC2 hit dispersion under identical injection — full 5×3 truth grid (rows = p_y , cols = p_x , MeV/c)



Step 2 — From PDC Dispersion to Reco Momentum

Transfer function. The RK4 fitter inverts $\vec{p}_{\text{target}} \mapsto (\vec{r}_{\text{PDC1}}, \vec{r}_{\text{PDC2}})$. Under the $\sigma_{\text{wire}} \sim 0.2$ mm assumption that goes into the Fisher matrix, PDC1/PDC2 positions should map to $\sigma_p \sim$ few MeV/c — this is what the analyzer reports. With the Sn target disabled, the physical PDC σ (Step 1) drops to 3–15 mm — still $\sim 15\text{--}75\times$ larger than wire smearing, but not the $\sim 500\times$ seen with the target in.

Consequence. Taking the 50 same-truth replicas and measuring how the *reconstructed* momentum scatters (histogram below), the post-fix global G4/Fisher halfwidth ratios are $p_x : 0.81$, $p_y : 1.86$, $p_z : 0.80$, $|\vec{p}| : 0.79$. Fisher is slightly conservative on $|\vec{p}|$, p_x and p_z (ratios 0.79–0.81; realised scatter $\sim 20\%$ narrower than the error bar), and $\sim 1.9\times$ too optimistic for p_y . The entire “ p_x reconstruction bias” seen pre-fix was the missing lab→target rotation; after the 2026-04-19 fix the median p_x residual is $|\langle \Delta p_x \rangle| \lesssim 3$ MeV/c across the grid.

Frame. All $p_x/p_y/p_z$ below are in the target frame (see the “Coordinate frames” slide early in Physical Setup). Pre-fix PNGs had a spurious $\Delta p_x \approx -p_z \sin \alpha \approx -627 \times \sin 3^\circ \approx -33$ MeV/c from the lab/target mismatch.

Source scripts:

- `scripts/analysis/plot_reco_momentum_dispersion.py`

- Output:

`docs/reports/reconstruction/figures/g4_fluctuation/reco_momentum/px{. . .}_py{. . .}`

What the Fisher σ Band Does (and Does Not) See

Fisher input channels — everything it knows comes from the RK forward model:

$$F = J^T \Sigma_{\text{meas}}^{-1} J, \quad \Sigma_p^{\text{Fisher}} = F^{-1},$$

where $J = \partial(\vec{r}_{\text{PDC1}}, \vec{r}_{\text{PDC2}}) / \partial(p_x, p_y, p_z)$ is the RK Jacobian at the best-fit momentum and Σ_{meas} is the PDC wire-position covariance ($\sigma_{\text{wire}} \sim 0.2$ mm per layer). That is *all* the Fisher matrix has seen.

Geant4 physics that Fisher is blind to (every one of these *adds* spread that shows up in the histograms):

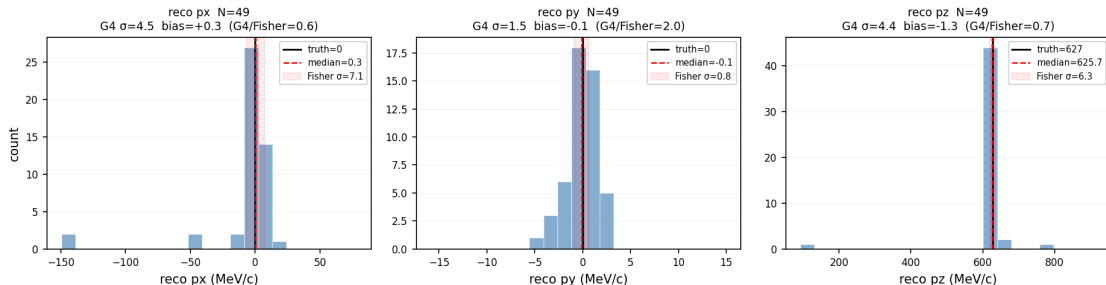
- multiple Coulomb scattering in PDC gas, window foils, ~ 1 m of air between PDC1 and PDC2;
- ionisation-energy-loss straggling (Landau/Vavilov) — only the *mean* dE/dx enters the RK propagator;
- δ -electron emission, secondary nuclear interactions;
- PDC drift-gas cluster-diffusion tails beyond the Gaussian σ_{wire} assumption.

Ensemble histograms are the other knob — they contain everything. Fixing truth (p_x, p_y, p_z) and running 50 independent Geant4 replicas (different seeds) samples the full stochastic chain; the histogram width is the realised σ_p^{G4} .

Reading the G4/Fisher ratio:

Reco Momentum Dispersion — Baseline (0, 0, 627)

Reco momentum dispersion: identical Geant4 injection (truth p_x, p_y, p_z) = (+0, +0, 627) MeV/c, N=49

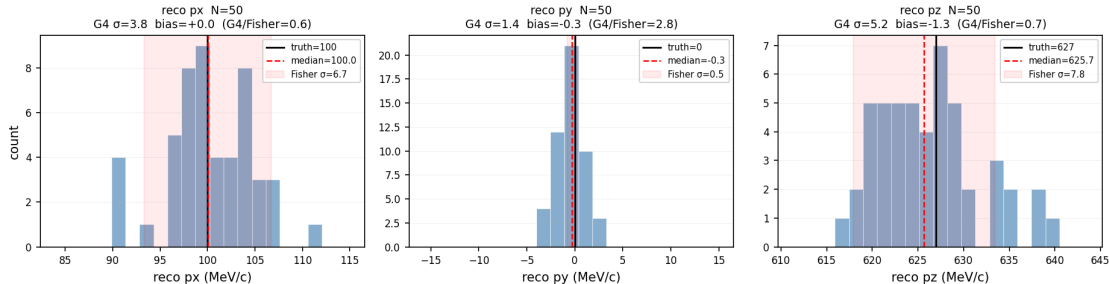


Black line = truth, red dashed = G4 median, pink band = Fisher σ around median. $N=49$ replicas.

Post-fix numbers (target frame). p_x : median at +0.3 MeV/c (was -33 pre-fix, purely the coordinate-frame artifact), G4 $\sigma=4.5$ vs. Fisher $\sigma=7.1$, ratio 0.63 — Fisher slightly conservative. p_y : ratio 1.96 (G4 1.5 vs. Fisher 0.8), the one direction where Fisher under-reports the realised spread. p_z : ratio 0.69 (G4 4.4 vs. Fisher 6.3), median residual -1.3 MeV/c. The only remaining issues are p_y under-sizing and (u, v, p) linearisation residuals.

Reco Momentum Dispersion — (+100, 0, 627)

Reco momentum dispersion: identical Geant4 injection (truth p_x, p_y, p_z) = (+100, +0, 627) MeV/c, N=50



N=50. Post-fix (target frame). p_x : median at +0.03 MeV/c (centered), G4 $\sigma=3.8$ vs. Fisher $\sigma=6.7$, ratio 0.57 (Fisher conservative). p_y : ratio 2.78 (G4 1.4 vs. Fisher 0.5) — this is the largest mismatch across the 15 truth points (Fisher tightest, G4 typical). p_z : ratio 0.67 (G4 5.2 vs. Fisher 7.8). Point-by-point ratio heterogeneity is real, $\sim 2\text{--}3\times$ across the grid; summarised on the next slide.

PDC → Reco Amplification — Summary

Matching the PDC-plane spread (Step 1) with the reco-momentum spread (Step 2) per truth point (source: `pdc_dispersion_summary.csv`, `reco_momentum_dispersion_summary.csv`):

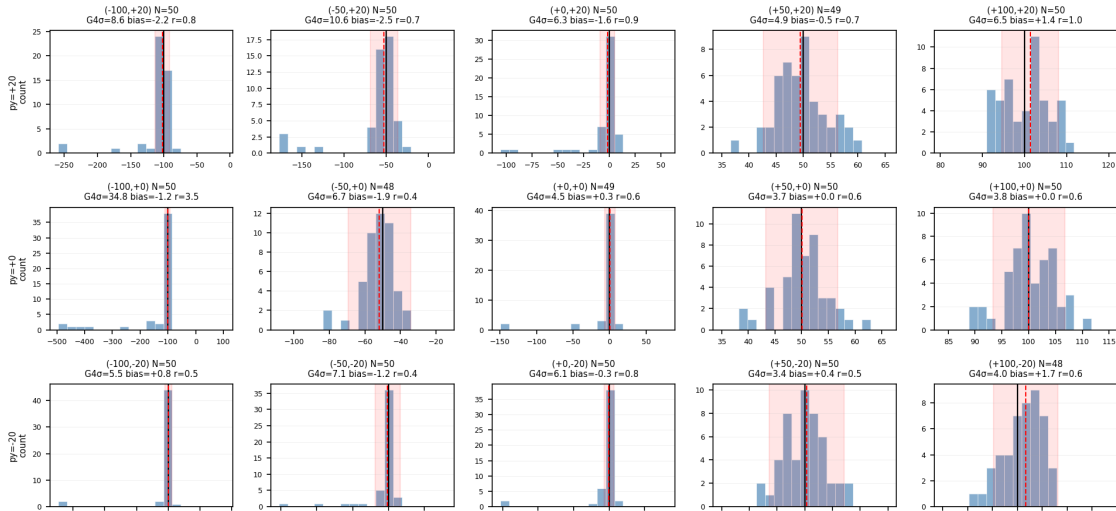
Truth		PDC dispersion (mm)				G4/Fisher ratio		
p_x	p_y	σ_x^1	σ_y^1	σ_x^2	σ_y^2	p_x	p_y	p_z
0	0	2.8	11.2	3.5	15.2	0.63	1.96	0.69
+100	0	3.7	9.1	5.7	12.3	0.57	2.78	0.67
-100	0	2.7	10.7	3.0	14.3	3.51	1.63	3.51
0	+20	2.9	9.6	4.0	12.6	0.89	1.90	0.72
0	-20	3.2	8.4	4.6	10.6	0.84	1.61	0.73

Read-across:

- Without the Sn target, PDC σ_y collapses from ~ 100 mm to ~ 10 – 15 mm and the reco-momentum G4/Fisher ratios sit at 0.6–0.9 for p_x, p_z and 1.5–3 for p_y — Fisher *sizing* is essentially correct everywhere except p_y .
- $(-100, 0)$ is the one point where p_x/p_z G4 halfwidths blow up to ~ 35 and ~ 22 MeV/ c (ratio 3.5). This is the LM coarse-grid coverage hole at the PDC acceptance edge (inward-bending trajectories with marginal geometry); see coverage findings for RK at that grid point in the project memo.

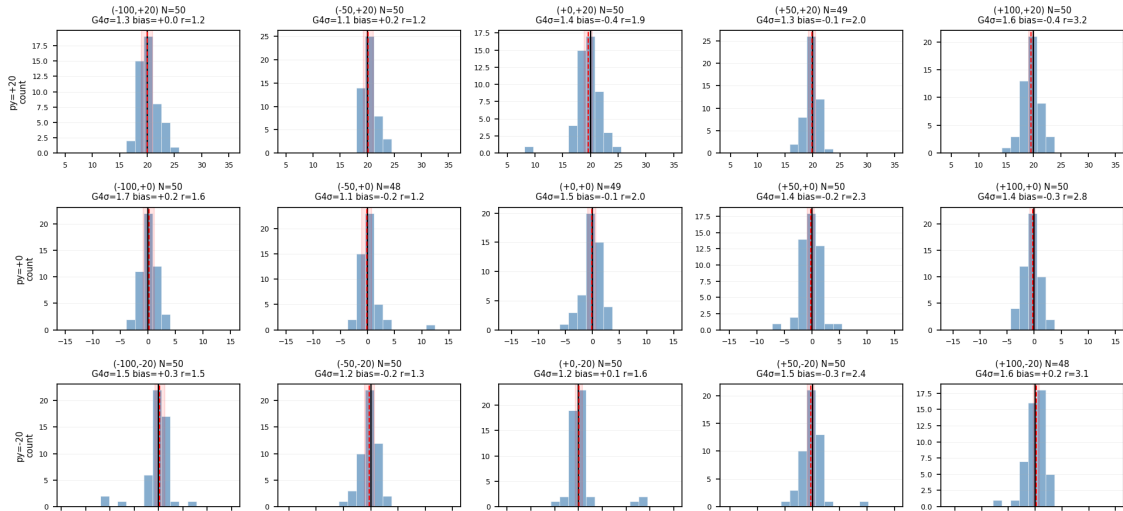
Reco p_x Dispersion — Full 5×3 Truth Grid

Reco p_x dispersion — full 5×3 truth grid (rows = p_y , cols = p_x , MeV/c). Black = truth, red dashed = G4 median, pink = Fisher σ band.



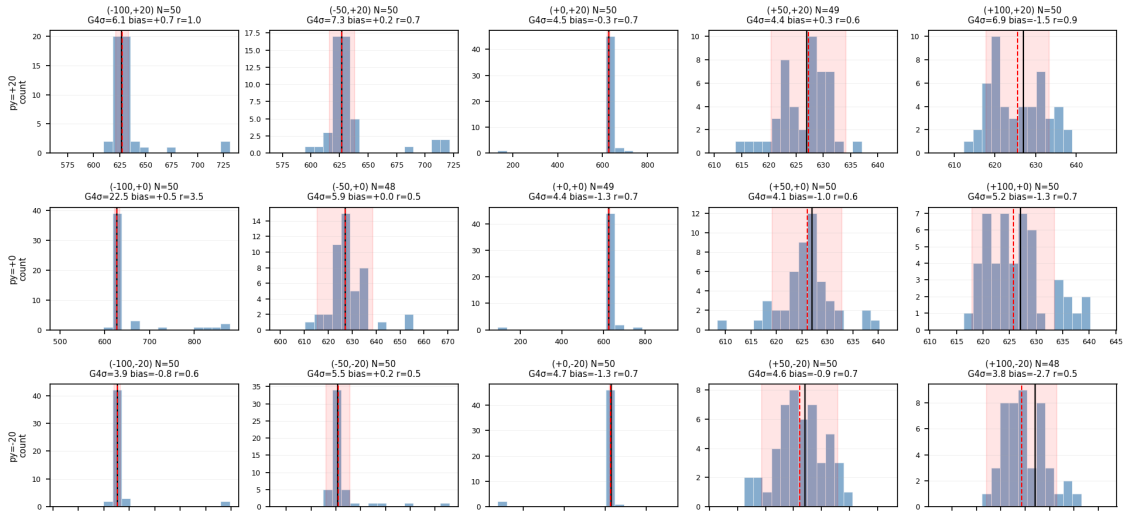
Reco p_y Dispersion — Full 5×3 Truth Grid

Reco p_y dispersion — full 5×3 truth grid (rows = p_y , cols = p_x , MeV/c). Black = truth, red dashed = G4 median, pink = Fisher σ band.



Reco p_z Dispersion — Full 5×3 Truth Grid

Reco p_z dispersion — full 5×3 truth grid (rows = p_y , cols = p_x , MeV/c). Black = truth, red dashed = G4 median, pink = Fisher σ band.



Summary

- ➊ **RK trajectory model:** relativistic RK4 integration of the Lorentz equation in the SAMURAI field map. Parameterized by ($u = p_x/p_z$, $v = p_y/p_z$, $p = |\vec{p}|$).
- ➋ **Residual model:** 2D wire-coordinate projection with Cholesky whitening. 4 independent residuals from 2 PDC planes, $N_{\text{dof}} = 1$.
- ➌ **Optimization:** Levenberg-Marquardt with coarse grid search ($7 \times 3 \times 5 = 105$ candidates) and multi-start from top 5.
- ➍ **Uncertainty:** Fisher (fast), Profile Likelihood (asymmetric), Laplace posterior, MCMC, and MC nonlinear propagation (256 samples).
- ➎ **Performance** (from E2E test): RK MAE = (20.4, 16.0, 10.8) MeV/c for (p_x, p_y, p_z); NN MAE = (21.0, 15.6, 3.1) MeV/c [NN from polluted model, see audit 2026-04-20; clean NN much better on p_x, p_y , similar on p_z]. Fisher $\sigma_{|\vec{p}|} = 2.4\text{--}6.0$ MeV/c.

- **dE/dx energy loss:** add Bethe-Bloch term to the RK4 propagator. Expected to remove the systematic -5 to -16 MeV/ c p_z bias.
- **Multiple scattering:** include Coulomb scattering in gas/window materials as process noise in the trajectory model.
- **Fix under-coverage** (post 2026-04-19 rotation fix): Fisher 68% coverage p_x 80%, $|\vec{p}|$ 76%, p_z 77%, p_y 42%. Profile (N=128): p_x 71%, $|\vec{p}|$ 78%, p_z 70%, p_y 37%. MCMC $|\vec{p}|$ 63%. p_x jumped from 5% $\rightarrow$ 80% just by rotating reco output to the target frame — the apparent -33 MeV/ c “bias” was a coordinate artifact, not a forward-model defect. The one remaining algorithm-level issue is p_y Fisher σ deflation by $\sim 2\times$ (p_y invariant under the rotation). Fix is a Cartesian (p_x, p_y, p_z) refit (no (u, v, p) linearisation) plus dE/dx in the RK propagator; retest on the 89k-event dataset once the fit is Cartesian.
- **Enable quality gate:** once validated, set `require_gate_pass=1` in E2E integration tests.